

DTVAD:End-to-End Autonomous Vehicle Planning Using Reinforcement Learning

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Abstract: Reinforcement learning (RL) based end-to-end autonomous driving faces significant challenges due to the sim2real gap, primarily because simulation environments are often insufficiently realistic. However, collecting data from the real world is highly risky, as RL requires exploration of potentially dangerous behaviors to learn which situations should be avoided. Additionally, the loss function of RL may hinder the optimization of the perception module. In this paper, we propose DTVAD, an RL-based end-to-end autonomous driving framework that directly leverages real-world datasets, specifically the nuScenes dataset, and employs a decision transformer as the planning module. Notably, we randomly sample 4,096 trajectories as the action space and utilize the Fréchet distance to measure the similarity between trajectories. For each sample in the nuScenes dataset, we select one of the top 70 most similar trajectories as the action for the corresponding state. The discrepancy between the selected action and the recorded trajectory may introduce potential collisions, enabling the agent to learn which behaviors should be avoided. And we use Decision Transformer(DT) as our planning head. We evaluate the effectiveness of DTVAD on the nuScenes dataset through open-loop test.

Key Words: end-to-end autonomous driving, planning, reinforcement learning

1 Introduction

Building an autonomous driving system is challenging because of the need to design and integrate multiple modules. Traditional autonomous vehicle systems often follow a pipeline paradigm, which makes it difficult to optimize the entire system due to information loss and the accumulation of errors.

In recent years, the end-to-end approach to autonomous driving has gained significant attention as a viable research avenue[1]. This method consolidates multiple modules and tasks into a unified model, demonstrating remarkable effectiveness in both benchmark datasets and simulated environments. By adopting this integrated framework, the entire system can be jointly optimized via backpropagation, enhancing overall performance.

However, end-to-end autonomous driving seems to be dominated by imitation learning(IL). Many works[1–5], have shown the effective of imitation learning. In RL area, some works[6, 7], use ground truth perception or map information, some model free reinforcement learning work[8] use simple perception. We argue that the reasons prevent reinforcement learning popular in end-to-end autonomous driving is that training RL agent or collecting data for RL in real world is dangerous, the gap between simulation and real world is hard to overcome, make RL agent is hard to be deployed in real world.

VADv2[9] generates a probabilistic distribution over potential trajectories, achieving remarkable performance in autonomous driving tasks. The generation of such distri-

butions is inherently compatible with RL. In this paper, we present DTVAD, a transformer-based RL method for autonomous driving, which use decision transformer[10] as planning head. We also training our model directly using real world data by adding reward to Nuscenes dataset.

Inspired by VADv2[9], We randomly sample 4096 trajectories from nuScenes[11] dataset as out trajectory vocabulary. The key thought of our work is we measure the similarity of the recorded trajectories in nuScenes[11] dataset and the trajectory in trajectory vocabulary by using fréchet distance. For every sample in nuScenes dataset[11], we choose one trajectory from trajectory vocabulary according to Fréchet distance as action. We can get the ego vehicle and other vehicles or objects in every sample, reward can be designed according to these information. Following[2, 3], we use a backbone to get bevfeature and use bevfeature to interact with prediction and map information, then combined with command embedding as state. We use decision transformer[10] as our planning head, the input of planning head is the stack of state, action, reward-to-go with time embedding.

Our contribution are summarized as follows:

- We proposed DTVAD, a reinforcement learning based end-to-end autonomous driving method, which can compared with imitation learning method.
- We proposed use fréchet distance to measure the similarity of two trajectory and convert the nuScenes[11] dataset to a offline reinforcement learning dataset.
- DTVAD demonstrates promising results on nusenes[11] dataset.

2 Related Work

2.1 End to End Autonomous Driving

Contemporary autonomous driving frameworks typically segment the operational workflow into three core components: perception, prediction, and planning. The evolution of online mapping technologies[12–14], coupled with

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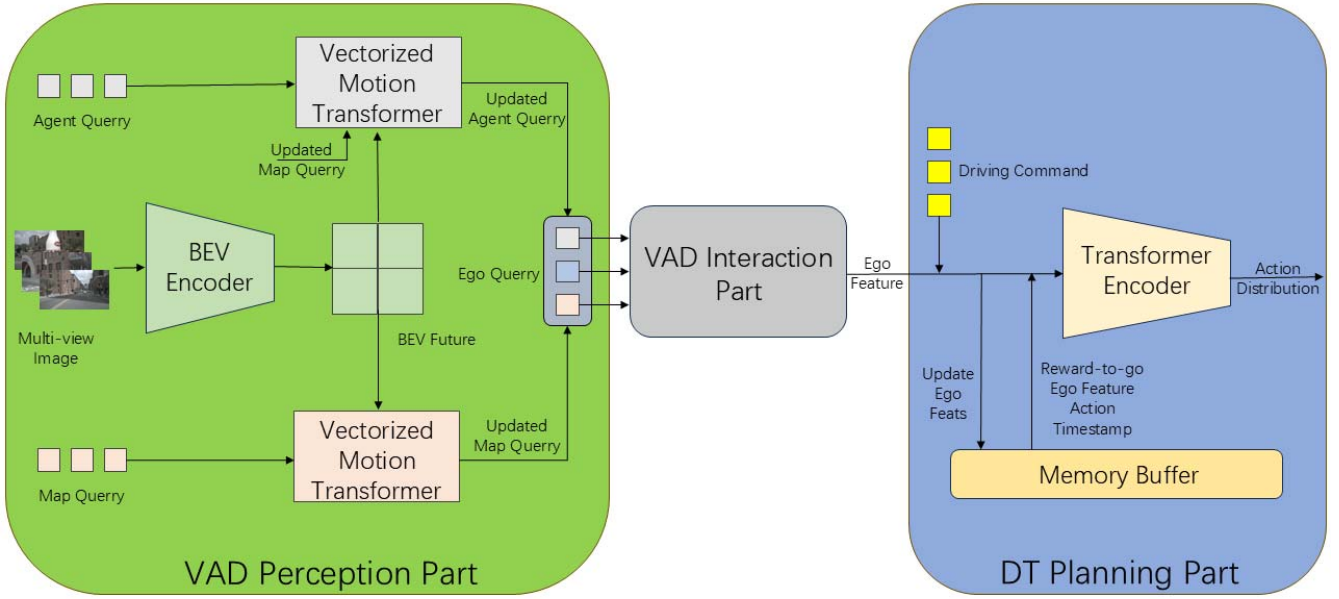


Fig. 1: The network architecture of DTVAD

advancements in integrated motion prediction[15–17], and perception[18–21], has catalyzed the development of numerous holistic autonomous driving approaches. UniAD[2] exemplifies this trend by strategically integrating multi-task information to facilitate goal-driven planning, demonstrating exceptional capabilities across perception, prediction, and planning domains. VAD[3] introduces an innovative paradigm by representing driving scenarios through comprehensive vectorized models, thereby eliminating the need for computationally demanding raster-based representations and manual post-processing procedures. VADv2[9] generating probabilistic distribution of trajectories and subsequently selecting one trajectory. However, RL significantly lagged behind both the modular pipeline approach and end-to-end imitation learning in terms of performance within the CARLA[22] simulator[23].

2.2 Transformer Based Reinforcement Learning

Transformer[24] has gained extensive application across diverse areas including natural language processing (NLP) and computer vision (CV)[25]. In the field of deep reinforcement learning (DRL), there's an emerging inclination towards integrating Transformers[26]. Initially, the incorporation of Transformers in RL sought to supplant conventional neural networks like CNNs and LSTMs. However, the direct deployment of the standard Transformer framework within RL decision-making frameworks has been fraught with practical stability issues[26]. Diverging from traditional RL methodologies that focus on estimating value functions or calculating policy gradients[27], Decision Transformers(DT)[10] employ a causal transformer framework to model a policy conditioned on rewards. Parallel research adopts a comparable strategy, utilizing Transformers to forecast environmental dynamics[28], Furuta et al. [29]

have extended the capabilities of DT[10] to match given hindsight information. This study explores the application of DT[10] in the context of autonomous driving systems, aiming to enhance decision-making processes in end-to-end self-driving scenarios.

3 Method

We present DTVAD, a Reinforcement Learning based end-to-end autonomous driving method. The overall framework of DTVAD is depicted in Fig.1. DTVAD takes 4096 trajectories from the nuScenes[11] dataset and designs a reward to encourage the ego vehicle to go ahead and avoid collision. Every sample in the nuScenes[11] dataset gets reward information.

The input of DTVAD is a multi-view image sequence, following[3], DTVAD first encodes the image features into BEV features[18, 19], and then DTVAD employs a set of agent queries and map queries to construct the vectorized scene representation. The main difference with VAD[3] is in the planning phase, inspired by DT[10], we use a transformer-encoder based planning head to project the stack of environment state, reward-to-go, action to the distribution of trajectories in the trajectory vocabulary.

3.1 Dataset And Reward

With the successful application of reinforcement learning in various domains[30–32], numerous studies have attempted to leverage reinforcement learning in the field of autonomous driving[6–8, 33, 34]. Reinforcement learning based methods mainly use data collected from simulation environments. However, the gap between the real world and simulation environments limits the applicability of reinforcement learning methods in real-world scenarios. We argue that the main reason preventing the use of real world data in RL methods is: (a) Collecting data from real world for

Table 1: Open-loop Performance Result

method	L2(m)				Collision(%)			
	1s	2s	3s	Avg.	1s	2s	3s	Avg.
ST-P3[5]	1.33	2.11	2.90	2.11	0.23	0.62	1.27	0.71
UniAD[2]	0.48	0.96	1.65	1.03	0.05	0.17	0.71	0.31
VAD-Tiny[3]	0.46	0.76	1.12	0.78	0.21	0.35	0.58	0.38
VAD-Base[3]	0.41	0.70	1.05	0.72	0.07	0.17	0.41	0.22
DTVAD-const-rtg	0.63	1.02	1.49	1.05	0.32	0.57	0.89	0.59
DTVAD-gt-rtg	0.43	0.65	0.95	0.68	0.45	0.60	0.79	0.61

Table 2: Ablation Study on Reward

Go Ahead Reward	Collision Reward	L2(m)				Collision(%)			
		1s	2s	3s	Avg.	1s	2s	3s	Avg.
✓		0.95	1.59	2.36	1.63	0.40	0.88	1.45	0.91
	✓	0.58	0.92	1.32	0.94	0.67	0.95	1.06	0.89
✓	✓	0.62	1.00	1.48	1.03	0.45	0.66	1.06	0.72

RL is dangerous because RL need wrong behavior, for example collision, to learn what scenaior should be avoid. (b) The real world datasets don't contain action and reward. We randomly sample trajectory from nuScenes[11] dataset and design reward to eccourage ego vehicle go ahead and avoid collision, which makes nuScenes[11] dataset to a offline reinforcement learning dataset.

- **Action Space.** Some previous work[8, 34], have utilized control commands, such as throttle and steering angle, as outputs. However, directly using control commands as outputs often results in limited interpretability and generalization[7]. So, We use the future trajectory of the ego vehicle as the output. Specifically, we sampled 4,096 trajectories from the nuScenes dataset to construct the action space, which is called trajectory vocabulary[9], including 1,024 trajectories for left-turning, 1,024 trajectories for right-turning, and 2,048 trajectories for straight driving. Different with[9], which selects the longest 4,096 trajectories, we emphasize maintaining the diversity of trajectories, thus we adopted a random sampling strategy.
- **Action and Reward.** The nuScenes[11] dataset do not contain reward information. In this paper, for every sample in nusenes dataset, the action is choosen form the 70 trajectories from trajectory vocabulary most similar to the ground truth trajectories. If none of the 70 trajectories would collide with other vehicle, the most similar trajectory will be adopt, otherwise the collision trajectory which most similar to the ground truth will be adopt. This ensures that when the ego vehicle follows the action trajectory, the surrounding environment is as similar as possible to the one recorded in the dataset. Deviations between trajectories could lead to potential collision risks, allowing the agent to learn which behaviors are dangerous. The designed reward is to en-

courage the ego vehicle go ahead and avoid collision, inspired by[6], the reward of DTVAD including two aspect: $r_{goAhead}$: The ego vehicle get reward of 1 for every 10-meter increment of the trajectory.

$r_{collision}$: The ego vehicle get reward of -5 when collision occur.

$$r = r_{goAhead} + r_{collision} \quad (1)$$

3.2 Network Architecture

We directly use the network in VAD[3] and DT[10] with little modify. The network architecture is not the importance of this paper, instead we focus on the dataset processing and the effectiveness of DT[10] as planning head.

- **Perception.** DTVAD use the perception part of VAD[3], The only differnet with VAD[3] perception part is that we convert the raw navigation signals into three learnable embeddings[2], this embedding is added to the ego state. This is because VAD[3] generate 3 trajectory for turn left, turn right and keep forward and choose one of them according to the navigation signal, but DTVAD predict only one action index, it is necessary that adding the navigation signals into ego state.
- **Planning.** We modify DT[10] as our palnning head. Specially, we use transformer encoder instaed of transformer decoder. In DT[10], the input contain a sequence of state at k timestamp. However, the perception part is computation intensive, a sequence of state will take too much time to make the training time or inference speed unacceptable. So, we add a memory buffer to store the states in last epoch and update states in current epoch.

3.3 Training and Inference

Although DTVAD is reinforcement-learning based, the training of DTVAD is end-to-end. The loss function can be

divide into perception loss and planning loss. Perception loss follows the vectorized scene learning loss in VAD, i.e. L_{map} and L_{mot} . As for planning loss, we use the corssentropy loss between the predicted action and the ground truth.

$$L = w_1 L_{map} + w_2 L_{mot} + w_3 L_{planning} \quad (2)$$

4 Experiment

In this section, we investigate the performance of DTVAD in the open-loop testing compared with imitation learning method. Following previous work[3, 5], displacement error(DE) and collision rate(CR) are employed to evaluate the planning performance.

4.1 Implement Details

The perception part of DTVAD used the perception part VAD-tiny[3]. We randomly sample 4096 trajectories from nusenes dataset as our trajectory vocabulary. We test two mode of DTVAD on nusenes[11] dataset open-loop testing. DTVAD-const-rtg set the reward-to-go as a const value at the beginning of every scene, and then reward-to-go will subtract the reward get in every sample. DTVAD-gt-rtg use the ground truth reward-to-go pre-computing from nusenes dataset, the purpose of DTVAD-gt-rtg is to show the performance of DTVAD when DTVAD get the correct reward-to-go information.

We use diffnet rtg and action information when comparing with other method and ablation study. The action is chosen from the 5 trajectories from trajectory vocabulary most similar to the ground truth trajectories when comparing with other method, this could reduce the likelihood of collision during action execution, thereby enhancing the algorithm's performance. In the ablation study, The action is chosen from the 5 trajectories from trajectory vocabulary most similar to the ground truth trajectories. This distinction is only present during the inference phase. The rtg and action information are same when training.

4.2 Open-loop Planning Result

Table 1 shows the performance of DTVAD on open-loop testing. Both DTVAD-const-rtg and DTVAD-gt-rtg performed far better than ST-P3. DTVAD-gt-rtg outperforms UniAD on DE and is comparable to VAD on DE. The open-loop test result shows DTVAD could be comparable even outperform IL method on DE. The performance of DTVAD on CR is inferior to that of IL method. Improving the performance of DTVAD on CR is a key focus for future work.

4.3 Ablation Study

Our ablation experiment focuses on the reward design. Table 2 shows the effectiveness of the go ahead reward and the collision reward. Go ahead reward encourages ego vehicle to go further and collision reward encourages ego vehicle to avoid collision. If go ahead reward was dropped, the ego vehicle may not go ahead to prevent the collision. If collision reward was dropped, the ego vehicle couldn't get the information that avoid collision needed. As shown in Table 2, compared with reward containing both go ahead reward and collision reward, the reward only containing go ahead reward performance worse in DE and CR, reward only containing collision reward performed slightly better in DE, but showed

a more significant decrease in performance in CR. CR is the more critical safety-related metrics. So, dropping any of go ahead reward and collision reward will cause worse performance, both go ahead reward and collision reward are necessary.

5 Conclusion

In this paper, we investigate the application of reinforcement learning(RL) in end-to-end autonomous driving using the nuScenes real-world dataset. We propose DTVAD, an RL-based approach that leverages the Decision Transformer architecture. Our method achieves excellent performance on the nuScenes dataset.

We hope this paper can inspire further research on reinforcement-learning based end-to-end autonomous driving with real world data. In addition to planning a safe and comfortable trajectory, accurately tracking the planned trajectory is equally important for autonomous vehicles. Therefore, intelligent control of autonomous vehicles is a direction worthy of research in the future[35–39].

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